Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/31/2026
PI Name:	Patla, Bijunath Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	303-497-4082
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City:	Boulder	**State:**	CO
Zip Code:	80305	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	04/01/2021	**End Date:**	09/29/2028
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: Grant/Contract number updated per D. Griffin/NASA-HQ (Ed., 1/22/25). NOTE: End date changed to 09/29/2028 per D. Griffin/NASA-HQ (Ed., 1/20/25).		
Key Personnel Changes/Previous PI:	New Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla; previous PI was Dr. Chris Oates.		
COI Name (Institution):			
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Task Description:

NOTE: Continuation of "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Oates)" with new PI, Dr. Bijunath R. Patla.
To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10^-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10^-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10^-17 or better.

• Test of gravitation redshift at 2 x 10^-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10^-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10^-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10^-16 or better.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 9 x 10^-19.

• Two ytterbium lattice optical clocks with uncertainties better than 2 x 10^-18.

• Strontium lattice optical clocks with uncertainties of 2 x 10^-18 (at JILA). A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons with demonstrated instability below 10^-16.

• A world-class time scale comprising eight hydrogen masers and a collection of Cs clocks.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time.

Gap 6 Mo. (2018): Develop and install precision time links between the MWL hardware site and NIST clocks. $80k for partial support of staff time.

Gap 12 Mo. (2018-2019): Assuming successful launch in 2020, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Develop a frequency-based time scale for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2019-2020): Assuming successful launch in 2021, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Evaluate and improve the performance of the new frequency-based time scale optimized for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2020-2021): Evaluate and improve the performance of the new frequency-based time scale. Upgrade hardware infrastructure to add resiliency and robustness to the frequency-based time scale. $80k for partial support of staff time.

Gap 13 Mo. (2021-2022):Develop and integrate methods for precise orbit determination. Accurately estimate

	higher-order relativistic effects,estimate time and frequency transfer noise for clocks in orbit. $90k for partial support of staff time. Gap 12 Mo (2022-2023) Assess relevant issues such as the adequacy of knowledge of the geopotential (including dynamical effects) for supporting world-wide time comparisons with advanced clocks. $80k for partial support of staff time. Prepare for the public release of data products in collaboration with ACES/ESA and extend investigations to benefit future missions with more precise clocks for testing fundamental physics. $93k for partial support of staff time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such as search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.
Task Progress:	The work has focused broadly on two main tasks: 1) ACES Microwave Link (MWL) terminal setup and testing at NIST; and 2) Clock infrastructure upgrades, data processing, time and frequency transfer from ground to space. 1) ACES Microwave Link (MWL) terminal setup and testing at NIST The microwave link (MWL) terminal arrived at NIST during the first week of August 2025. The European Space Agency (ESA) and Airbus crew were onsite following a series of meetings to plan the installation. The MWL terminal hosting platform is on the roof of the new building~81 (B81) at the NIST site. We installed three FSJ2-50 coaxial cables to provide a high-stability transmission medium for 100 MHz reference distribution, specifically engineered to minimize measurement uncertainty. Its solid-wall corrugated copper construction offers superior phase stability and a shielding effectiveness exceeding 120 dB, effectively isolating the reference signal from electromagnetic interference (EMI) and environmental thermal drift. This mechanical and electrical rigidity ensures that signal phase and group delay remain constant, maintaining the integrity of the primary frequency standard throughout the laboratory infrastructure. All cables were characterized, and two were assigned for transporting a 100 MHz clock signal and a synchronized 1 pulse-per-second (PPS) input feeding the MWL. The third one is used to transport the 1PPS generated from the 100MHz received at the MWL. The geographic coordinates (latitude, longitude, and altitude) of the ground terminal, obtained from Global Navigation Satellite System (GNSS) data, obtained by pre-positioning the antenna at the location of the MWL before installation. The coordinates are inputs used to calculate the precise angular relationships among the antenna, the geostationary satellite, and the Sun. The reference 100~MHz signal to the MWL terminal is the direct output of a Vector Atomic iodine vapor cell clock. While the Active Hydrogen Maser remains the primary standard for laboratory-grade frequency synthesis due to its unmatched stability for a few hundred seconds, the Vector Atomic Iodine Vapor Cell Clock's reliance on a molecular optical transition provides an inherently stable reference that is less susceptible to the magnetic and thermal sensitivities that can affect Maser performance. For ACES Microwave Link (MWL) operations, SFTP access at NIST is configured using a defense-in-depth approach that combines chroot confinement (SFTP “jail”) with strict firewall-based network isolation. External connections are granted via SFTP-only accounts confined to dedicated chroot environments, ensuring users can access only authorized data directories and cannot traverse the broader NIST filesystem. Interactive shells are disabled, internal-sftp is enforced, and chroot root directories are owned by privileged users to prevent escape from the confined environment. At the network layer, the secure file transfer standard (SFTP) service is protected by the NIST perimeter firewall with tightly scoped rules: inbound access is restricted to approved MWL source IP addresses and required ports only, while outbound connectivity from the SFTP host is minimized or blocked. The server resides on an isolated subnet/VLAN, preventing lateral movement into operational timing infrastructure. The connection to the MWL endpoint was tested and verified to be operating as intended, using controlled authentication and file-transfer trials. End-to-end SFTP sessions were successfully established through the firewall from approved source addresses, confirming proper enforcement of chroot confinement, SFTP-only access, and network isolation. Test transfers demonstrated reliable data ingress, confirming the configuration meets ACES MWL operational

and security requirements.

2) Clock infrastructure upgrades, data processing, time and frequency transfer from ground to space

We have also improved the uncertainty estimates for the Al ion and Yb lattice clocks to low E-19 and E-20 in fractional frequency. The clocks are also operating more often with high uptimes per run.

Currently, NIST operates three Yb-lattice clocks, including a portable clock with fractional frequency performance in the range of E-16 - E-18, a high-performance Al-ion clock, and Sr clocks from JILA. The Sr lattice clock at JILA has been operating at 50% uptime for ~100 days. There's also a second clock comparison campaign underway in Boulder to estimate frequency ratios using a shared Si cavity as a local oscillator, with an instability of E-16.

Aside from the introduction of VA0021 clock into the ensemble, the microstepper units were repaired and recently reintroduced as part of a broader infrastructure upgrade to support advanced timescale development and testing. In parallel with restoring hardware functionality, we have expanded the software framework to accommodate additional algorithms beyond the existing NIST operational timescale methods. These enhancements include the integration of a Kalman-filter–based estimator and a frequency-scale algorithm, enabling side-by-side evaluation with current approaches. Together, these repairs, infrastructure improvements, and software extensions provide a flexible testbed for validating new timescale strategies and assessing performance under realistic operational conditions as real-time clock data becomes available in the coming months. For tests of fundamental physics using the ACES clock network, we are tailoring theAshby-Patla clock comparison formalism, which provides a unified relativistic framework for time and frequency transfer between ground-based and spaceborne clocks by expressing all observables in a common coordinate time while rigorously accounting for gravitational potential differences, kinematic effects, and signal-propagation delays. This approach treats proper times at each clock as realizations of an underlying coordinate time and derives explicit relations for accumulated offsets and fractional frequency shifts along realistic worldlines. For clocks onboard the International Space Station, the formalism naturally incorporates Earth’s gravity field, orbital motion, and light-time corrections to enable high-accuracy comparisons between terrestrial clocks and the ISS clock, separating secular relativistic rate differences from periodic orbital signatures. Our formalism maps directly onto two-way time-transfer measurements and frequency links, yielding implementable expressions that support nanosecond-level synchronization and E-16 level frequency comparisons, while remaining extensible to more complex environments such as cislunar space and planetary missions.

More recently, to add more resiliency, the radio station clocks at Fort Collins are also compared with NIST timescales in Boulder in real time using upgraded multi-constellation GNSS receivers with live monitoring.

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